DR. PETAR N. PETROV

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RESEARCH APPOINTMENTS & EDUCATION

2025-Present **Institute of Science and Technology Austria (ISTA)**
Postdoctoral Visiting Scientist, Division of Mathematical & Physical Sciences

2020-Present **University of California, Berkeley**
Postdoctoral Scholar, Department of Physics
Advisor: Holger Müller

2014-2020 **Stanford University**
Ph.D. in Physical Chemistry
Advisor: W. E. Moerner
Thesis: *Improving the Precision and Accuracy of Three-Dimensional Single-Molecule Localization Microscopy*

2010-2013 **University of California, Berkeley**
B.S. in Chemistry with high honors
Research Advisor: Peidong Yang

Summer 2011 **Berkeley Center for Green Chemistry**
Research Intern
Advisors: Martin Mulvihill and Michelle Douskey

2007-2010 **University of Central Oklahoma**
High School Student Researcher, Department of Chemistry
Advisor: John Bowen

RESEARCH INTERESTS

Method development in optical & electron microscopy & spectroscopy; structure & spatio-temporal dynamics of biomolecules & their environments; theory & experiment in optical metrology; signal & image processing algorithms

GRANTS, FELLOWSHIPS, AWARDS, & HONORS

2023-2024 Ruth L. Kirschstein Postdoctoral Fellowship (F32 GM149186); NIH/National Institute of General Medical Sciences
2017-2020 Stanford Interdisciplinary Graduate Fellowship; Stanford University
2019 Poster Award, Stanford Bio-X Interdisciplinary Initiatives Seed Grants Symposium
2017,18,19 Bio-X Travel Award; Stanford University
2017,18 Second Best Poster; Stanford Photonics Retreat
2016 Best Poster; Stanford Photonics Retreat
2014 Glenn T. Seaborg Award for Outstanding Undergraduate Research in Chemistry; UC Berkeley
2011-2014 Melvin J. Heger-Horst Undergraduate Fellowship; College of Chemistry, UC Berkeley
2013 Undergraduate Research Stipend; College of Chemistry, UC Berkeley

TEACHING, MENTORSHIP, & OUTREACH

Teaching:

Fall 2016 Head Teaching Assistant; Chemical Principles Accelerated (Chem 31X); Stanford University
Fall 2015 Teaching Assistant; Chemical Principles Accelerated (Chem 31X); Stanford University
Winter 2015 Teaching Assistant; Physical Chemistry Laboratory (Chem 176); Stanford University
Fall 2014 Teaching Assistant; Chemical Principles Accelerated (Chem 31X); Stanford University
2013-2014 Independent Tutor (Mathematics, Chemistry)

Fall 2011 Co-Facilitator; Chemistry Undergraduate Internships & Research (Chem 98/198); UC Berkeley

Mentorship & Outreach:

2025 Stanford Bio-X Summer Undergraduate Summer Research Program
Created and led workshops for undergraduate students on scientific oral and poster presentations

2022 The Compass Project at Berkeley
Provided one-on-one mentorship to an undergraduate physics student at UC Berkeley

2021 Be A Scientist (Bay Area Scientists In Schools)
Led middle school students in developing, conducting, and presenting science experiments in their classroom

2018-2019 Stanford Splash
Created and taught short courses on microscopy for high school students

2017 Stanford Science Penpals
Corresponded by mail with high school students from low-income backgrounds about science, college, and graduate school

2017 Inspiring Future Scientists Through Shadowing
Supervised and mentored a high school student in a research project on white-light interferometry

2016 Stanford Summer Undergraduate Research Fellowship Program
Trained and mentored a visiting undergraduate student in an independent research project on adaptive optics

SERVICE

2018-Present Ad-hoc reviewer
Nature Communications, Optica, Optics Express, Biomedical Optics Express, Optics Letters

2011-2014 Berkeley Undergraduate Chemical Society; Co-Founder & Webmaster

2012 Berkeley Chemical Review; Founding Editor

SELECTED PRESENTATIONS

Invited Talks:

12. Molecular Foundry Symposium on Free Electron Quantum Physics; Berkeley, CA, USA; August 2024
11. From Solid State to Biophysics; Cavtat, Croatia; June 2024
10. TU Wien; Vienna, Austria; May 2024
9. Instruct Image Processing Center, Seminar on Image Processing; virtual; October 2023
8. Gordon Research Seminar on Three Dimensional Electron Microscopy; Newry, ME, USA; June 2023
7. Rosalind Franklin Institute; Didcot, UK; April 2023
6. Laboratory of Molecular Biology; Cambridge, UK; April 2023
5. One World Cryo-EM Seminar Series; virtual; March 2023
4. Institute for Pure and Applied Mathematics; Los Angeles, CA, USA; November 2022
3. University of Vienna; Vienna, Austria; September 2022
2. Max Planck Institute for the Science of Light; Erlangen, Germany; September 2022
1. Max Planck Institute of Molecular Cell Biology and Genetics; Dresden, Germany; September 2022

Other Talks:

13. Vienna Region Cryo-EM Seminar; Vienna, Austria; April 2026
12. University of Vienna, Modern Concepts in Structural Biology Seminar; Vienna, Austria; January 2026
11. ISTA, CryoCoffee Seminar; Klosterneuburg, Austria; January 2026
10. UC Berkeley, Compass Lecture; Berkeley, CA, USA; March 2025
9. ISTA; Klosterneuburg, Austria; May 2024
8. Bay Area Cryo-EM Meeting; Redwood City, CA, USA; October 2023
7. International Microscopy Congress; Busan, South Korea; September 2023

6. Quantitative BioImaging; Oxford, UK; January 2020
5. Quantitative BioImaging; Rennes, France; January 2019
4. Quantitative BioImaging; Göttingen, Germany; January 2018
3. Stanford University, Single-Molecule Seminar Series; Stanford, CA, USA; February 2017
2. SPIE BiOS, Single Molecule Spectroscopy and Superresolution Imaging Conference; San Francisco, CA, USA; January 2017 [doi]
1. Pentasectional Meeting of the American Chemical Society; Norman, OK, USA; April 2010

Posters:

16. Gordon Research Conference on 3D Electron Microscopy; Barcelona, Spain; June 2026
15. Austrian Cryo-EM Symposium; Klosterneuburg, Austria; November 2025
14. Frontiers in Electron Microscopy for Physical and Life Sciences; Princeton, NJ, USA; October 2024
13. Gordon Research Conference on 3D Electron Microscopy; Newry, ME, USA; June 2023
12. eBEAM School on Nano-Optics with Free Electrons; Porquerolles, France; September 2022
11. Stanford Bio-X Interdisciplinary Initiatives Seed Grants Symposium; Stanford, CA, USA; February 2020
10. Stanford Bio-X Fellowship Symposium; Stanford, CA, USA; October 2019
9. Stanford University Photonics Retreat; Marshall, CA, USA; April 2019
8. Stanford Bio-X Interdisciplinary Initiatives Seed Grants Symposium; Stanford, CA, USA; February 2019
   · *Received Poster Award*
7. Stanford Bio-X Symposium; Stanford, CA, USA; August 2018
6. Stanford Photonics Retreat; Pacific Grove, CA, USA; April 2018
   · *Awarded Second Best Poster*
5. Stanford Bio-X Fellowship Symposium; Stanford, CA, USA; October 2017
4. NIH Common Fund 4D Nucleome Annual Meeting; Bethesda, MD, USA; September 2017
3. Stanford University Photonics Retreat; Pacific Grove, CA, USA; April 2017
   · *Awarded Second Best Poster*
2. Stanford University Photonics Retreat; Marshall, CA, USA; April 2016
   · *Awarded Best Poster*
1. UC Berkeley Saegebarth Undergraduate Research Fair; Berkeley, CA, USA; April 2013

PUBLICATIONS

(* indicates co-first author; † indicates corresponding author)

Pre-Prints:

1. Y Yu*, A Cheng*, E Montabana, N Paz Soldán, ES Cooper, JT Zhang, JJ Axelrod, **PN Petrov**, A Torkaman, D Roof, M Derstine, B Buijsse, W Hagen, D Kimanius, S Zheng, M Kopylov, M Paraan, D Balakrishnan, J Hutchings, H Cheng, J Remis, A Singh, L Maisenbacher, CS Potter, H Müller, B Carragher, D Agard, PK Olshin, "A Crossed Laser Phase Plate for CryoEM," bioRxiv [doi]

Peer-Reviewed Papers:

19. JJ Axelrod, L Maisenbacher, A Singh, IM Pope, **PN Petrov**, JT Zhang, H Müller, "A thermoelastic limit on the focal intensity in Fabry-Pérot cavities," *Phys. Rev. A* **114**, 043507 (2026) [doi]
18. **PN Petrov***, JT Zhang*, J Remis, JJ Axelrod, H Cheng, ES Cooper, IK Hicklin, S Sandhaus, C Schnurr, RM Glaeser, H Müller, "Laser phase plate improves structure determination of small proteins by cryo-EM," *Science* **393**(6807), 195-199 (2026) [doi]
17. **PN Petrov**, JT Zhang, JJ Axelrod, PK Olshin, H Müller, "Crossed laser phase plates for transmission electron microscopy," *Nat. Commun.* **17**, 7199 (2026) [doi]
16. J Remis, **PN Petrov**, JT Zhang, JJ Axelrod, H Cheng, S Sandhaus, H Müller, RM Glaeser, "Cryo-EM phase-plate images reveal unexpected levels of apparent specimen damage," *J. Struct. Biol.* **216**(4), 108150 (2024) [doi]

15. JJ Axelrod, JT Zhang, **PN Petrov**, RM Glaeser, H Müller, "Modern approaches to improving phase contrast electron microscopy," *Curr. Opin. Struc. Biol.* **86**, 102805 (2024) [doi]
14. JJ Axelrod, **PN Petrov**, JT Zhang, J Remis, B Buijsse, RM Glaeser, H Müller, "Overcoming resolution loss due to thermal magnetic field fluctuations from phase plates in transmission electron microscopy," *Ultramicroscopy* **249**, 113730 (2023) [doi]
13. A-K Gustavsson, RP Ghosh, **PN Petrov**, JT Liphardt, WE Moerner, "Fast and parallel nanoscale three-dimensional tracking of heterogeneous mammalian chromatin dynamics," *Mol. Biol. Cell* **33**(6), 1-11 (2022) [doi]
· *Selected as a Highlight from MBoC*
12. **PN Petrov**, H Müller, RM Glaeser, "Perspective: Emerging strategies for determining atomic-resolution structures of macromolecular complexes within cells," *J. Struct. Biol.* **214**(1), 107827 (2022) [doi]
11. C Turnbaugh, JJ Axelrod, SL Campbell, JY Dioquino, **PN Petrov**, J Remis, O Schwartz, Z Yu, Y Cheng, RM Glaeser, H Müller, "High-power near-concentric Fabry-Perot cavity for phase contrast electron microscopy," *Rev. Sci. Instrum.* **92**, 053005 (2021) [doi]
· *Selected as Editor's pick*
10. **PN Petrov**, WE Moerner, "Addressing systematic errors in axial distance measurements in single-emitter localization microscopy," *Opt. Express* **28**(13), 18616-18632 (2020) [doi]
9. HW Bennett*, A-K Gustavsson*, CA Bayas, **PN Petrov**, N Mooney, WE Moerner, PK Jackson, "Novel fibrillar structure in the inversin compartment of primary cilia revealed by 3D single-molecule super-resolution microscopy," *Mol. Biol. Cell* **31**(7), 619-639 (2020) [doi]
· *Selected as a Highlight from MBoC*
8. L Möckl, AR Roy, **PN Petrov**, WE Moerner, "BGnet: Accurate and rapid background estimation in single-molecule localization microscopy with deep neural nets," *Proc. Natl. Acad. Sci. U.S.A.* **117**(1), 60-67 (2020) [doi]
7. L Möckl, **PN Petrov**, WE Moerner, "Accurate phase retrieval of complex 3D point spread functions with deep residual neural networks," *Appl. Phys. Lett.* **115**, 251106 (2019) [doi]
· *Selected as Editor's pick*
6. A-K Gustavsson, **PN Petrov**, WE Moerner, "Light sheet approaches for improved precision in 3D localization-based super-resolution imaging in mammalian cells," *Opt. Express* **26**(10), 13122-13147 (2018) [doi]
5. A-K Gustavsson, **PN Petrov**, MY Lee, Y Shechtman, WE Moerner, "3D single-molecule super-resolution microsocopy with a tilted light sheet," *Nat. Commun.* **9**, 123 (2018) [doi]
4. Y Shechtman, A-K Gustavsson, **PN Petrov**, E Dultz, MY Lee, K Weis, WE Moerner, "Observation of live chromatin dynamics in cells via 3D localization microscopy using Tetrapod point spread functions," *Biomed. Opt. Express* **8**(12), 5735-5748 (2017) [doi]
3. **PN Petrov**, Y Shechtman, WE Moerner, "Measurement-based estimation of global pupil functions in 3D localization microscopy," *Opt. Express* **25**(7), 7945-7959 (2017) [doi]
2. MP Backlund, A Arbabi, **PN Petrov**, E Arbabi, A Faraon, WE Moerner, "Removing orientation-induced localization biases in single-molecule microscopy using a broadband metasurface mask," *Nat. Photonics* **10**(7), 459-462 (2016) [doi]
1. A Fu, H Gao, **PN Petrov**, P Yang, "Widely Tuneable Distributed Bragg Reflectors Integrated into Nanowire Waveguides," *Nano Lett.* **15**(10), 6909-6913 (2015) [doi]

Software:

3. `tm2d`: high-resolution 2D template matching for cryo-EM; associated publication forthcoming
2. `eer2tiff`: dose fractionation for cold field emission guns; see publication 18
1. `easy-pupil-finder`: phase retrieval for 3D localization microscopy; see publication 3